Griffiths 2021 Breast Cancer — analysis summary

RUN 3 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample integrates 74 samples across two technologies. Mixing is judged on the per-neighbourhood iLISI (eval-spec metric), median 1.02→11.71 — not on the coarse same-sample-neighbour scalar. The manifold resolves into a large Body (91.8k) and a low-complexity left lobe; the Body reproduces the reference Gast↔pSom cross-repression and is one continuous mass (single kNN-graph component; 96% coherent diffusion-map mass) → GO, scoped to the Body.

Gate 2 — module geography → GO (non-branching)

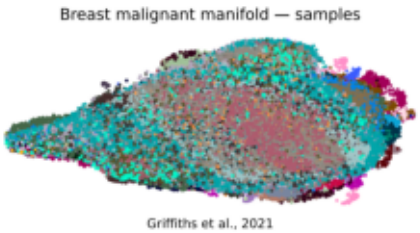
Organized Gast/pSom cross-repression (smoothed r=−0.45). Along pseudotime pSom falls (ρ=−0.65) while Crest rises (ρ=+0.57); NC-high decile at t=0.85 (terminal). Joint pSom&Crest enrichment 0.31<1 — inverse terminal gradient, a single (Gast/pSom)→Crest axis. FLAG: Crest co-enriches the Gast pole marginally more than pSom (0.33 vs 0.31) — arm inversion flagged, pass anyway.

Gate 3 — hub cross-repression → NA

Non-branching topology (principal curve: 2 tips, 0 forks across N=15/20/30; fold-free self-approach 2.86). No bifurcation to test → NA (branching-only correlation/modules notebooks not produced).

STEP FIGURES

Manifold — samples



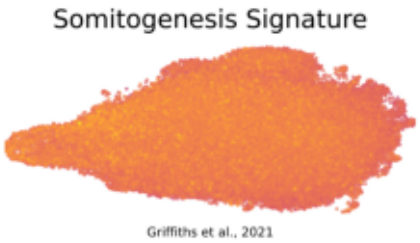
Pseudotime



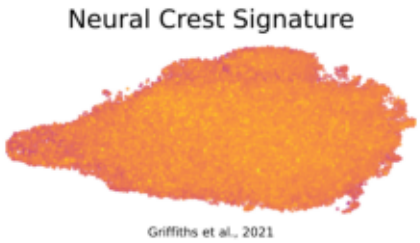
Gastrulation



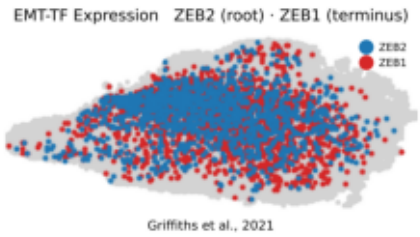
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (ER+ neoadjuvant) (Griffiths et al., 2021), 10x + iCell8 — 91,802 malignant cells analysed (Body compartment; 74 samples ≥50).

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching), with organized Gast/pSom cross-repression (r=−0.45) and a terminal Crest gradient; no Gast↔pSom bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the other CCCA breast datasets. Gate 1 GO / Gate 2 GO / Gate 3 NA. Raw module means Gast −0.020 / pSom −0.053 / Crest +0.016. EMT-TFs ZEB2 (early/root) → ZEB1 (late/terminus).

Parameter choices

Threshold	≥50 malignant cells/sample (no dominant count cliff) → 77/95 samples, 110,999 cells; Body = 91,802	**Compartment**	Scoped to the Body lobe; separated left lobe (16,545) is a low-complexity technical partition (NEAT1/MALAT1-led) excluded
Embedding	Harmony θ=10 on sample (removes 10x/iCell8 batch); n_pcs=30, n_neighbors=30, spread=1.3; 4,885 overdispersed; no sc.pp.scale	**Trajectory**	scFates curve (non-branching), Nodes=20 λ=0.3 μ=100; fold-free (SA 2.86); corr=+0.94
Root	Gast/pSom-high, Crest-low tip; axis not proliferation-driven (pt vs cycling ρ=+0.07)	**Geometry**	Linear; raw means Gast −0.020 / pSom −0.053 / Crest +0.016
EMT-TFs	ZEB2 (early/root) → ZEB1 (late/terminus); ZEB1 9.3% / ZEB2 5.3%		